Ex No: 10

1.9.26

Simulation of RIP using Cisco Packet Tracer.

AIM:

To simulate Routing information Protocol (RIP) using CISCO Packet Tracer and observe dynamic routing between different networks.

IP configuration:

Device	Interface	IP Address
PC0	Fast Ethernet	10.0.0.218
Router 0	Fa0/1	10.0.0.1/8
Router 0	S0/0/1	192.168.1.254.30
Router 1	S0/0/0	192.168.1.250/30
Router 1	S0/0/1	192.168.1.246/30
Router 2	S0/0/0	192.168.1.245/30
Router 2	S0/0/1	192.168.1.253/30
PC1	Fasth Ethernet	20.0.0.2/8

Procedure:

1) Create the topology with 3 routers and 2 PCs.

2) Connect the routers using serial cables and connect PC's to the respective routers.

3) Assign IP addresses to all PCs and router interface.

4) Enable the rower interfaces.

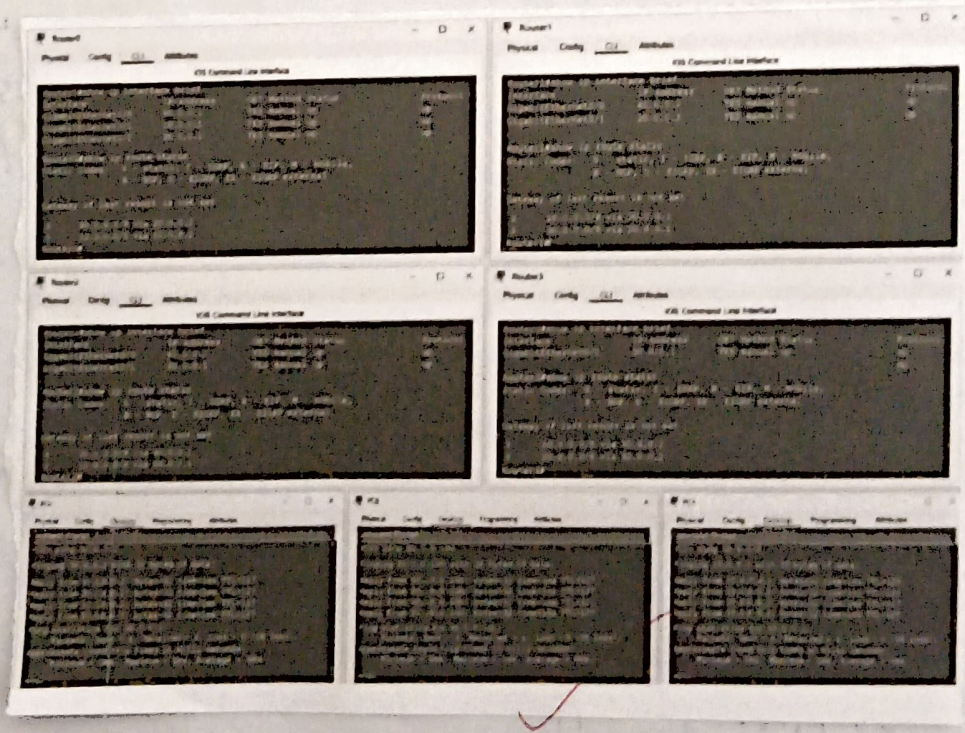
5. Configure the remaining interfaces according to the given IP table.

6. Enable RIP on Router 0 and configure RIP on Router 1 and 2.

7. From PC0, use ping 20.0.0.2 to verify connectivity.

8. Use tracert to observe the route taken by packets.

9. Disconnect the direct Router 0 - Router 2 link and observe that RIP reroutes the traffic through Router 1.



PC0 — Command Prompt (ping)

```
C:\> ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126
Reply from 20.0.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\> tracert 20.0.0.2
Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1     1 ms      1 ms      1 ms      10.0.0.1
  2     2 ms      2 ms      2 ms      192.168.1.253
  3     3 ms      3 ms      3 ms      20.0.0.2

Trace complete.
```

Result:

Thus, RIP was successfully simulated in Cisco Packet Tracer, and dynamic route selection and routing were observed.